Autonomous Charging Network — One-Page Service Level Agreement & Operating Specification

Autonomous Energy Certainty · Outcome-based Charging-as-a-Service (CaaS) · Project Sentinel

Contract Form No: KEMP-AUTO-2029-SLA-V2 · **Jurisdiction:** Republic of Finland / EU Energy Law · **Effective:** January 1, 2029 · **Parties:** Kempower Autonomous Energy Services Oy (Service Provider) · Lahden Seudun Liikenne (Lahti City Transit) & European Logistics Depot Alliance (Client)

1 · What is promised — Guaranteed SLOs

SLO	Gold target	Titanium tier upgrade · breach trigger
SLO-1 Fleet on-time full-charge rate	≥ 99.50 %	≥ 99.90 % (≤ 3 missed trips/yr) · monthly on-time rate below target
SLO-2 Dispenser fleet availability	≥ 99.50 %	≥ 99.90 % (≤ 43 min down/mo) · minute-level $\Sigma T_{avail}/\Sigma T_{total}$ over 720 h
SLO-3 Islanded operational resilience	≥ 72 h	Full 1,425 kW local DLM · unplanned stall or grid-limit breach during WAN loss
SLO-4 Grid-limit compliance	100.0 %	Absolute — incl. 95.5 % rectifier buffer · any instantaneous transformer breach
SLO-5 Self-healing control latency	≤ 100 ms	≤ 50 ms (<10 ms logic + BBM shift) · response latency > 200 ms

Silver tier baselines: 99.00 % / 99.00 % (≤ 432 min) / 24 h / 100 % / ≤ 200 ms. **Carve-outs from the measurement window** (clock suspends, bar never lowers): unannounced medium-voltage outages or DSO hard curtailment at non-BESS sites, vehicle-side BMS faults, MRDT-short dwell, sabotage. Announced curtailment (e.g. G1) stays inside the promise.

2 · What the system decides — Autonomous authority

Operational scenario	Algorithm direct authority	Mandatory human escalation
1 · Dynamic power routing & virtual replugging	Full — reallocates 25 kW DPR blocks on departure countdown & SoC deficit	Human intervention prohibited
2 · Spot-price spikes & grid curtailment (G1–G3)	Full — pre-charges flexible loads, derates low-urgency stalls (0.0 kW in G1)	Only a mandatory DSO grid cut-off order
3 · Module thermal anomaly & hardware faults	Full — millisecond BBM isolation; N-1 power pooling sustains operation	Work order auto-logged; technician dispatched in day
4 · Insulation / grounding breakdown (IMD)	Full — contactor trip, gun isolated, connector locked out	Field crew after 3 failed retries
5 · Total WAN loss (network outage)	Full — local edge takes over, 1,500 kW local DLM	Telecom provider if outage > 72 h
6 · Vehicle crash, fire, emergency stop	Full — master substation breaker opens within 10 ms	Mandatory on-site inspection & certified manual reset

Hard physical safety trips — firmware-locked, no software or cloud override: cabinet ≥ 65 °C → linear derate + reverse fan pulse; ≥ 75 °C → hardware trip, master AC contactor opens ≤ 100 ms; connector pin ≥ 90 °C → DC disconnect ≤ 50 ms; insulation < 100 Ω/V → IMD fault, HV closure inhibited; residual DC ≥ 30 mA → RCD trip ≤ 40 ms (IEC 61851-23); P_{grid} > 1.00 × transformer limit → overload clamp on non-urgent modules ≤ 20 ms.

3 · What it costs — Fees, penalties & the physics gate

250 €	5.00 €/min	1.50 €/kWh	≤ 150 %
base delayed-departure penalty (Ti 350 · Ag 150)	progressive delay after 15 min grace (7.50 / 3.00; 10 / 20 min)	missing-energy compensation (2.00 + rental / 1.00)	aggregate monthly liability cap of base fee

Fleet penalty (MRDT-compliant departures):

$$\text{Penalty} = \Sigma_{delayed}\,[\,P_{base} + \max(0,\, t_{delay} - 15\ \text{min})\cdot C_{min}\,] + \Sigma_{undercharged}\,[\,\Delta SoC/100 \cdot Cap_{batt} \cdot C_{energy}\,]$$

Availability fee deduction: 0 if ≥ 99.50 % · **2.0 % of monthly fee per 0.1 pp** shortfall (max 90 %) if 95.00–99.50 % · **100 % waiver** if < 95.00 % (service default trip).

MRDT physics gate (moral-hazard shield): before any credit is paid, the minimum physically required dwell time is computed per departure:

$$MRDT_j = (\,\Delta SoC_j/100 \cdot Cap_{batt,j}\,) \div (\,\min(P_{charger,max},\, P_{vehicle,max,j}) \cdot \eta_{system}{=}0.95\,) \times 60 + T_{buffer}{=}15\ \text{min}$$

Liquidated damages apply **if and only if** actual dwell ≥ $MRDT_j$. A late-arriving vehicle (dwell < MRDT) is charged best-effort and the provider is fully indemnified — the gate that kills the "arrived 10 min late with 200 kWh to go" claim.

Caps & billing integrity: fleet penalties per depot ≤ 100 % of its monthly fee; aggregate monthly payout ≤ **150 %** of base fee; invoices strictly non-negative (Net Payable = max(0, Fee − Credits), excess credits roll over); no indirect or consequential damages.

4 · Who is accountable when it decides wrong

Black-box evidence, no disputes: (1) edge gateway records voltage, current, pin temperature, fan RPM, CAN/BMS frames at **1-second resolution** on NVMe with **72 h** power-loss WAL; (2) telemetry streams emit periodic **SHA-256 Merkle** root hashes anchored to cloud ledgers in real time or ≤ 15 min after WAN restoration; (3) after any anomaly, the offline discrete-event simulator performs automated root-cause attribution within **15 minutes** of reconnection and verified liquidated damages are **auto-credited on the next invoice**.

Provider is relieved of liability only upon objective black-box proof of: utility blackout / unannounced DSO curtailment (BESS sites beyond the 2 h design buffer; non-BESS sites S1–S3, S5 fully indemnified from the moment of occurrence, hours excluded from SLO-1/2); vehicle BMS non-compliance or internal battery faults (ISO 15118 diagnostic dumps); client MRDT non-compliance; external mechanical sabotage or impact.

Kept human by design: the system never overrides a firmware safety interlock; hardware faults that need hands (73 dispatches = 5.09 FTE) are retained as costed technician service inside the fee — software does not turn wrenches, and the zero-miss busbar path that is absent from the official price list is handed back unpriced rather than invented.

5 · Execution & verification seal

Party A — Service Provider	Party B — Contracted Client
Kempower Autonomous Energy Services Oy Signatory: Kempower System Chief Architect	Lahden Seudun Liikenne (Lahti City Transit) & European Logistics Depot Alliance Signatory: Pekka Virtanen, Fleet Operations Director

Complete contractual summary. Full tier tables, derivations, 2029 customer story & equations: SLA_Specification_Annex_Full.pdf · Every figure traces to the unmodified official dataset (45 CSVs / 240,635 rows), re-verified by verify_all.py.
Execution date: January 1, 2029 · Smart-contract root hash: 0x8F9C4A12E8B907D3E2F4A5C6718293A4B5C6D7E8F90123456789ABCDEF012345
Verification gateway: https://sentinel.kempower.com/verify/contract/2029-LSL-DEPOT-01